

Jishnu Auro Ghosh

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Summary

Hybrid Industrial Engineer and Software Builder specializing in applied AI, optimization, and advanced manufacturing systems. Bridging the gap between physical operations and scalable software. Proven track record leading Python-based scheduling architecture for Boeing, building full-stack ERP dashboards, and deploying ML pipelines. Actively seeking high-impact roles in Deep Tech, AI Infrastructure, or Robotics.

Education & Certifications

Purdue University

B.S. in Industrial Engineering | Minor: Law & Society

Coursework: Operations Research, Linear Programming, Industrial Control Systems, Manufacturing Processes

West Lafayette, IN

Aug 2022–May 2026

Google

Google AI Professional Certificate

Comprehensive training in applied artificial intelligence, predictive modeling, and scalable ML workflows.

Online

2026

DeepLearning.AI

TensorFlow Developer Professional Certificate

Online

2025

Technical Skills

Programming & Dev:: Python, SQL, JavaScript, PHP, R, MATLAB, Git/GitHub, VS Code

AI, Data & Solvers:: TensorFlow, scikit-learn, pandas, NumPy, Gurobi, Tableau, Data Visualization

Optimization:: Linear Programming (Simplex, Interior Point), Operations Research, Stochastic Modeling

IE / Operations:: Production Scheduling, Capacity Analysis, Bottleneck Analysis, Facility Layout

Engineering, Analytics & Software Experience

Boeing 787 Production Capstone

Software Architect (5-Member Team)

West Lafayette, IN

Jan 2026–May 2026

- Led the technical development of a Python-based production scheduling engine for Boeing's 787 Final Assembly rollout.
- Translated complex shop-floor constraints (labor, physical layout, task readiness) into a scalable, automated decision-support model.
- Engineered clear, traceable KPIs and scenario planning workflows, successfully handing off a robust, enterprise-ready GitHub repository to Boeing operations stakeholders.

Human Integration Lab, Purdue

Research Assistant – AI & Biometrics

West Lafayette, IN

Feb 2026–Aug 2026

- Supported advanced human-systems research focusing on teamwork and AI-assisted workflows in VR construction environments.
- Engineered automated Python and Excel scoring pipelines to process over 160 hours of raw fNIRS, eye-tracking, and biometric sensor data into actionable performance metrics.

Enterprise Systems Dashboard (Purdue Operations)

Full-Stack Developer

West Lafayette, IN

Fall 2025

- Built a fully functional ERP/SCM dashboard from scratch using SQL, PHP, and JavaScript.
- Designed normalized data schemas and optimized queries to surface real-time operational metrics, enabling visibility into cross-functional workflows for senior-management decision support.

Margo Katherine Wilke Research

Data Analyst & Undergraduate Researcher

West Lafayette, IN

Aug 2024–Aug 2025

- Built longitudinal text datasets and reproducible Python analysis workflows using hypothesis testing and regression to extract actionable insights from large, unstructured data.

Selected Technical Projects

Optimization Algorithm Development

Algorithm Architecture & Implementation

Fall 2025

- Built highly scalable Simplex and Interior Point method solvers entirely from scratch in Python, implementing complex degeneracy and anti-cycling handling.
- Benchmarked custom algorithms directly against commercial solvers to validate runtime performance and accuracy.